Abdomen allometry, ovary development, and growth of female snow crab, *Chionoecetes opilio* (Brachyura, Majidae), in the northwestern Gulf of St. Lawrence

Marianne Alunno-Bruscia and Bernard Sainte-Marie

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Abstract: Growth and maturation of female snow crab (*Chionoecetes opilio*) in the northwestern Gulf of St. Lawrence were elucidated using carapace size-frequency distributions, indices of molting activity, ovarian mass, and laboratory and field measures of growth per molt. Females develop in three major stages: immature, with narrow abdomen and no detectable ovaries; prepubescent, with narrow abdomen and previtellogenic followed by vitellogenic ovaries; and adult, with broad abdomen and reproductive capability. Additionally, there is an ephemeral pubescent stage represented by females temporally close to the maturity molt. A reduction in the rate of carapace and abdomen growth occurs at the passage from immature to prepubescent, because energy is diverted into germinal growth, and adulthood is reached at a terminal molt to maturity. The pattern of abdomen growth relative to carapace is complex, consisting in successive phases of low, high, and again low positive allometry with increasing size of immature–prepubescent females. Over the period 1989–1996, a few females became adult at instar VIII at 4.5 years postlarval age, but more commonly, maturity occurred at instars IX and X at ages of 5.5 and 6.5 years, respectively. Average size at maturity may be temperature dependent, and within cohorts, larger females may mature earlier than smaller females.

Résumé : La croissance et la maturation de la femelle du crabe des neiges (*Chionoecetes opilio*) dans le nord-ouest du golfe du Saint-Laurent ont été élucidées à l'aide de distributions des tailles de la carapace, d'indices de mue, de la masse des ovaires et de mesures au laboratoire et en nature des accroissements de la taille à la mue. La femelle se développe en trois étapes principales: immature, avec un abdomen étroit et sans ovaires apparents; prépubère, avec un abdomen étroit et des ovaires d'abord prévitellogéniques puis vitellogéniques; et adulte, avec un abdomen élargi et une capacité de reproduction. De plus, il existe une étape éphémère appelée pubère que représentent les femelles qui effectueront bientôt leur mue de maturité. Le passage d'immature à prépubère s'accompagne d'une réduction du taux de croissance de la carapace et de l'abdomen, en raison d'une allocation d'énergie à la croissance des ovaires, et le passage à l'étape adulte se fait lors d'une mue terminale de maturité. La croissance de l'abdomen relativement à la carapace suit un schéma complexe; on reconnaît des phases successives de faible, de forte et à nouveau de faible allométrie positive, en fonction de la taille croissante des femelles immatures et prépubères. Au cours de la période 1989–1996, de rares femelles sont devenues adultes au stade de mue VIII à l'âge postlarvaire de 4,5 ans, mais le plus souvent la maturité survenait aux stades de mue IX et X à l'âge de 5,5 et 6,5 ans respectivement. La taille moyenne à la maturité serait dépendante de la température et la maturité serait atteinte plus tôt par les grandes que par les petites femelles d'une cohorte.

Introduction

The snow crab *Chionoecetes opilio* is a major fisheries resource in the Northwest Atlantic (FAO 1993). In eastern Canada, a variety of conservation measures were implemented early in the history of the fishery, including a minimum legal size of 95 mm carapace width (CW) and a prohibition against landing females (Miller 1976). Both measures effectively prevent female exploitation and presumably protect stock reproductive potential (Bailey and Elner 1989). Consequently, fisheries research for *C. opilio* initially tended to focus on males, even though detailed information on the biology of females may be critical to proper management of the resource. While research efforts over the past few years have filled large gaps in knowledge of the reproductive biology of female *C. opilio* (reviewed in Elner and Beninger 1992, 1995), information on growth and maturation phases for females in eastern Canada remains sketchy and rests on a few pioneering studies (e.g., Powles 1968; Watson 1970; Moriyasu et al. 1987; Robichaud et al. 1989).

In several majid crabs other than *C. opilio*, analysis of abdomen and (or) chela allometry for females has revealed three distinct developmental stages (e.g., Teissier 1933, 1935; Vernet-Cornubert 1958*a*; Hartnoll 1963; Aldrich 1974). Depending on author, the first is called an immature, juvenile, or nonpubescent stage; the second, an adolescent or prepubescent

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stage; and the third, a mature, adult, or pubescent stage. In all of these species the passage to the mature stage occurs at a terminal molt, and this is also thought to be the case for female *C. opilio* (e.g., Ito and Kobayashi 1967; Watson 1970). The two first stages in the development of secondary sexual characters each usually comprise at least two successive molt instars, although there are known exceptions with only one molt instar per stage (Rice and Hartnoll 1983). In female *C. opilio*, to date, only immature and mature developmental stages have been unequivocally identified by analysis of abdomen allometry (Watson 1970; Conan et al. 1990). However, based on various anatomical criteria, Lanteigne et al. (1996) categorized females as immature (with white ovaries), premature (with orange ovaries), and mature. The size ranges for immature and premature females were 19–60 and 24–68 mm CW, respectively, and ovary maturation proceeded over more than one molt cycle. To the contrary, Moriyasu et al. (1987) stated that full development of ovaries in *C. opilio* occurs within one intermolt period.

In brachyuran crabs, the onset of ovary development usually causes a reduction of body growth (e.g., Hartnoll 1982). For laboratory-held female *C. opilio*, Miller and Watson (1976) and Moriyasu et al. (1987) found a simple linear relationship between pre- and post-molt CW for females prior to the molt to maturity but a marked decrease in carapace incremental growth at the molt to maturity. This pattern is consistent with an abrupt (i.e., over one molt cycle) passage from an immature to a mature developmental stage, whereby massive energetic investment in ovaries is reflected in body growth only at the maturity molt (Moriyasu et al. 1987). However, the lower size limit of females in the two laboratory growth experiments was barely less than, or slightly in excess of, the size of the smallest females in which Lanteigne et al. (1996) detected developing ovaries. If immature females do require more than one molt cycle to develop ovaries, the possibility remains that in *C. opilio*, as in other brachyurans, there is a reduction in body growth at initiation of ovary development which has not been detected.

The objectives of the present study on *C. opilio* were to produce a comprehensive model of female growth and maturation, and to contrast developmental patterns for females and males. Our model for females is based on an intensive 1-year field study of size structure and ovary development for the Baie Sainte-Marguerite population (northwestern Gulf of St. Lawrence) and on a laboratory experiment to measure carapace and abdomen molt increments, both of which were conducted in 1991–1992. While preliminary analysis of these data by Alunno-Bruscia (1993) established several fundamental features of female growth and development, including age at maturity, some of the processes in the penultimate and ultimate molts could not be resolved for lack of sufficient numbers of large females in field samples. This situation was attributed to interannual fluctuation in the size and maturity structure of the *C. opilio* population. Further investigation has revealed that the *C. opilio* population in Baie Sainte-Marguerite is influenced by alternating waves and troughs of recruitment, which recur approximately every 8 years (Sainte-Marie and Dufour 1995; Sainte-Marie et al. 1996). A wave and trough are each composed of a few consecutive year-classes either moderate to strong or very weak. In the present study we have circumvented the problem of transient scarcity of large females by expanding our analysis to include some of the data from an 8-year monitoring program for *C. opilio* in Baie Sainte-Marguerite. The present contribution complements an earlier study on growth and maturation of male *C. opilio* (Sainte-Marie et al. 1995) and is part of a much larger effort to decipher patterns and processes for major life-history events of *C. opilio*.

Material and methods

Nomenclature

The terminology used in this study for the different types of females is the following. The "immature" category includes females in which no ovaries could be detected by naked eye. The "prepubescent" category groups females with ovaries easily seen by the naked eye and which are at the previtellogenic stages or early vitellogenic stages. "Pubescent" females have ovaries containing large yolk-laden oocytes and are temporally close (approximately 13 days or less) to the maturity molt, such that they induce persistent precopulatory grasping behavior in males (Watson 1970, 1972; Sainte-Marie and Hazel 1992; Bouchard et al. 1996). Among females that have undergone the molt to maturity, three categories may be recognized. "Nulliparous" females have not yet extruded eggs, "primiparous" females carry their first clutch of eggs, and "multiparous" females carry their second or subsequent clutch of eggs. Because the distinction between female types is not always easy or necessary, we also use the terms "juvenile" to designate collectively females that have not yet undergone the molt to maturity, and "adult" to designate females that have performed the maturity molt.

Field sites and collection of crabs

The sampling and sorting methods for this study are the same as those described in previous studies on *C. opilio* (Sainte-Marie 1993; Sainte-Marie et al. 1995, 1996), so we provide below only a brief summary of information pertaining to these aspects. The principal field site, Baie Sainte-Marguerite, is localized in the northwestern Gulf of St. Lawrence, eastern Canada (ca. 50°06′N, 66°35′W). Sampling was conducted in 1991 during 11–15 April, 19–24 June, 22–24 August, 23–29 October, and 11–12 December; in 1992 during 19–21 March and 12–14 May; in 1995 during 2–9 May; and in 1996 from 27 April to 2 May. On every occasion, randomly allocated positions were sampled in each of three depth strata: 0 to ≤20 m, >20 to ≤80 m, and >80 to 140 m. Crabs were collected with a 3-m beam trawl fitted with heavy tickler chains, to dislodge buried animals, and 15-mm netting in the codend to retain crabs exceeding ≈15 mm CW. Crabs <15 mm CW are retained once the mesh becomes occluded with animals and debris.

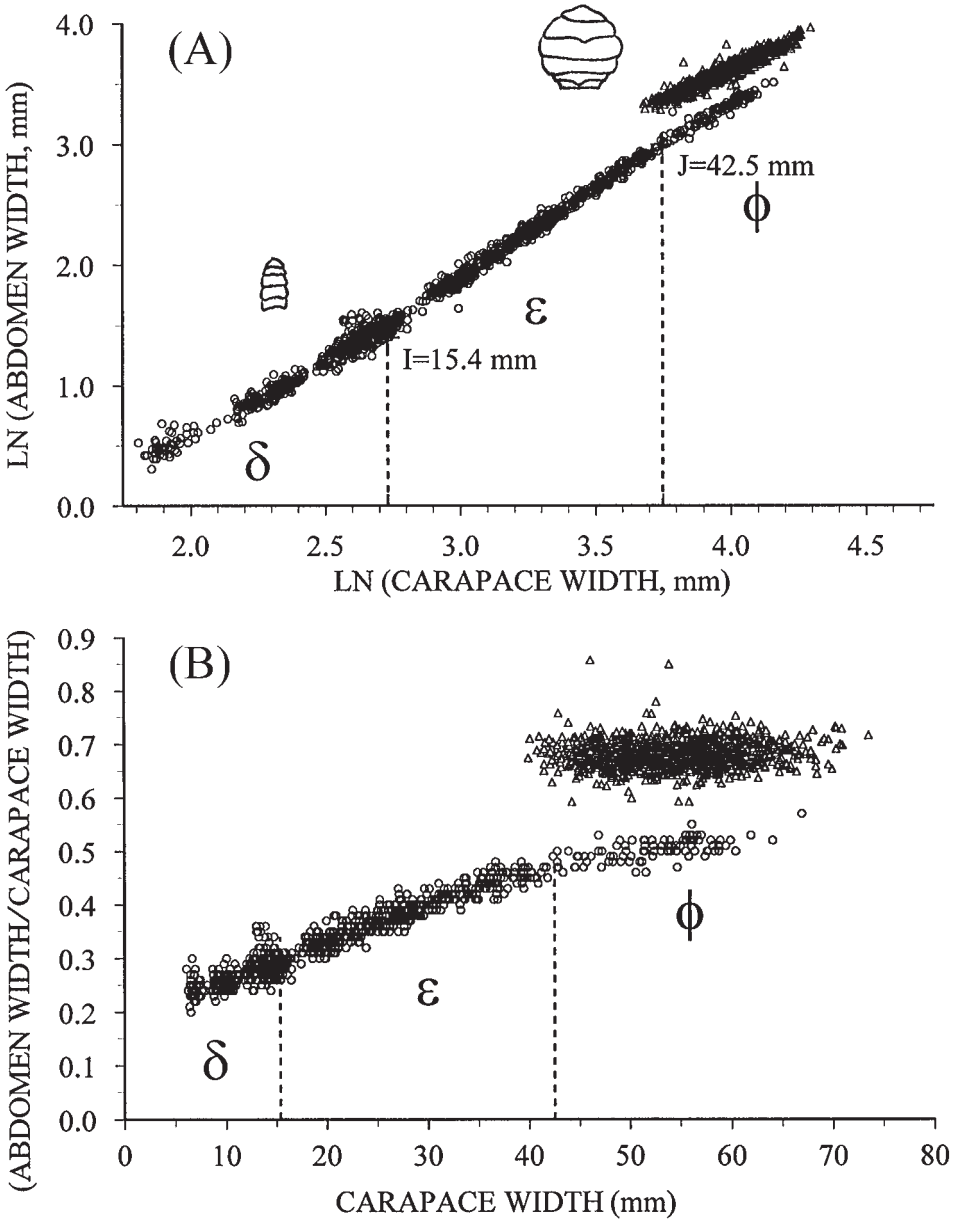
Additionally, on bottoms <35 m deep in Baie Sainte-Marguerite, SCUBA divers collected pubescent females from 19 to 29 March 1992 and pubescent mating pairs (i.e., a male clasping a pubescent female) from 19 to 24 March 1996. Juvenile female *C. opilio* for laboratory experiments were collected by beam trawl on the south shore of the Lower St. Lawrence Estuary (ca. 48°33′N, 68°35′W) from 14 to 28 September and from 19 to 23 October, 1992, at depths of 40–230 m. Live females were kept in 200-L shipboard tanks with running fresh seawater, until they molted or were transferred to holding facilities at Institut Maurice-Lamontagne. When molting occurred in shipboard tanks, the exuvium was removed and the soft-shelled crab was isolated for ≈1 day, so that its exoskeleton would harden somewhat, and then both the exuvium and crab were measured as described below.

Laboratory experiment

Molt increments for female *C. opilio* were determined on specimens held in captivity from October 1992 to April 1993. Over this period, females <34 mm CW were kept in 60-L tanks at densities <15 individuals per tank (i.e., <55 individuals·m^{-2}) while larger females

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Fig. 1. Juvenile (○, *N* = 1684) and adult (△, *N* = 1410) female *Chionoecetes opilio* collected in Baie Sainte-Marguerite from April 1991 to March 1992. (A) Scattergram of ln(abdomen width, AW; in mm) against ln(carapace width, CW; in mm). Three groups of juvenile (δ, ε, and ϕ) females can be determined by inflections in the line relating ln(AW) to ln(CW). Abdomen shape for females of 12 and 38 mm CW are illustrated. (B) Scattergram of the ratio of abdomen width to carapace width against carapace width.



were kept in 1700-L tanks at densities not exceeding 12 individuals·m^{-2}. Tanks were supplied with running seawater (10% fresh seawater renewal per day) averaging 1.5°C and 28.9‰, with extremes of –1.0 to 6.1°C and 24.7 to 32.4‰. Photoperiod was controlled to reflect natural light conditions. Crabs were fed previously frozen shrimp (*Pandalus borealis*) semiweekly and tanks were checked for molting crabs on a daily basis. When a crab molted, the exuvium was removed and the live crab was isolated for 2–3 days so that it could harden, and thereafter both the exuvium and live crab were measured as described below. Postmolt crabs were then fixed by injection of, and immersion in, 4% formalin in seawater, for later use in determining the extent of ovary development according to criteria given below.

Biological observations

Crabs were sexed on the basis of the shape of the abdomen. Small females have a subrectangular abdomen, while large females have an oval or rounded abdomen (Fig. 1A; see also Ito 1957) that contrasts with the triangular abdomen of males. However, crabs <6–7 mm CW could not be assigned to gender by visual examination alone, because the abdomens of females and males are not well differentiated, and therefore remained unsexed.

Exoskeleton condition was rated 1 (clean soft), 2 (clean hard), 3 (intermediate), 4 (dirty hard), or 5 (dirty soft) according to criteria in Sainte-Marie (1993). These ratings reflect progressive change and eventually deterioration of the carapace with time elapsed since the last molt (Sainte-Marie and Dufour 1995). On numerous occasions, complete (i.e., sexable) exuviae or molting females were collected by beam trawl. Molting females had swollen pericardial sacs and the old shell was raised, thereby revealing a new soft shell beneath.

Female crabs were categorized into juvenile or adult categories

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using the shape and relative size of the abdomen. Juvenile females have a flat abdomen, which is conspicuously narrower than the cephalothoracic sternites, while adult females have a bulging and broad abdomen that almost completely covers the cephalothoracic sternites. Adult females were further divided into nulliparous, primiparous, or multiparous categories based mainly on exoskeleton condition. Nulliparous and primiparous females have a clean and iridescent exoskeleton, which may be either soft or brittle due to the recent molt to maturity, or hard, but which is unspoiled (i.e., exoskeleton conditions 1 or 2) or only lightly scarred and fouled (i.e., less deteriorated stages of exoskeleton condition 3). However, nulliparous and primiparous females are distinguished by the fact that the former have not, while the latter have, extruded a clutch of eggs. Multiparous females are characterized by a more or less dirty and dull exoskeleton, which is usually heavily scarred (including grasping marks) and fouled and may be either hard (i.e., exoskeleton conditions 3 or 4) or soft due to decay (i.e., exoskeleton condition 5).

Using an unmodified vernier caliper (Sainte-Marie and Hazel 1992), widths of the carapace in its broadest part (CW) and of the fifth abdominal tergite taken across its middle (AW) were measured to the nearest 0.1 mm on all female exuviae and live females. To determine ovary mass and maturity, a sample of juvenile females from Baie Sainte-Marguerite was fixed by injection of, and immersion in, 4% formalin in seawater on each of the first six sampling periods of 1991–1992. The mass of blotted ovaries (OM) was determined to the nearest 0.01 mg and their color was recorded. Because ovaries could not be detected by naked eye or extracted in very small females, only those >28.9 mm CW were considered for quantitative analyses of ovary development.

As in the study on male growth (Sainte-Marie et al. 1995), three different indicators were used to reveal female molting periods from field data. The first was the temporal change in CW frequency distributions. The second was the frequency of occurrence by size of molting females, of female exuviae, and of soft-shelled females. The third indicator was the presence of females with a developing shell beneath the old shell.

Data analysis

Values of AW plotted against CW, after transformation to their natural logarithm, may assume a more or less curvilinear pattern. One approach to modeling this pattern is to use a polynomial regression model. Alternatively, the pattern can be modeled by sequential straight lines with different slopes following the method described by Somerton and MacIntosh (1983), which determines the intersection point between two convergent lines. The latter method facilitates interpretation of biological patterns: the abscissa value of an intersection point separates by size two different developmental stages, and the regression coefficients represent a measure of relative growth in the two stages (Hartnoll 1974, 1982).

A simple linear regression model was used to relate ln(OM) to ln(CW) on each of the sampling periods. Comparison of significant regressions among different periods proceeded first by testing slopes for equality by analysis of variance (ANOVA) and then by testing elevations of lines with homogeneous slopes for equality by analysis of covariance (ANCOVA) (Sokal and Rohlf 1981). Ovary mass adjusted to a constant CW by ANCOVA provided an index of gonad development which is independent of exoskeleton fouling and injuries that frequently modify female body mass (Sainte-Marie 1993).

The distribution of CWs for unsexed and female *C. opilio* collected by beam trawl was represented by histograms. Modal structure in histograms was analyzed with the MacDonald and Pitcher (1979) method. Successive modes were identified as instars, and between sampling periods a shift in abundance from one mode to the next was interpreted as a molting event (Hartnoll 1982). The Hiatt growth diagram (Hiatt 1948; Kurata 1962; Mauchline 1976) was used to model female carapace and abdomen growth at molt. The Hiatt growth model relates postmolt (L_{T+1}) to premolt (L_T) size by linear least-squares regression as $L_{T+1} = bL_T + a$. Crabs missing limbs in the premolt condition were excluded from analyses, because they may have smaller relative molt increments than intact crabs (Hiatt 1948; Bennett 1973; Miller and Watson 1976; Smith 1990). Analysis and comparisons of regressions for various groupings of females proceeded as described above for ovary growth models. The abscissa value of the intersection point (I) between two lines with significantly different slopes was calculated as $I = (a_2 - a_1)/(b_1 - b_2)$, where a_1 and a_2 are the intercepts and b_1 and b_2 the slopes of the two lines.

Unless specified otherwise, statistics presented in text are the mean and SE. Adjustment of regression models was verified when appropriate using a lack of fit test (Draper and Smith 1981).

Results

Abdomen allometry

In the scattergram of ln(AW) against ln(CW), juvenile females clustered along a sinuous line marked initially by positive curvature and then by negative curvature (Fig. 1A). The smallest juvenile females had a narrow subrectangular abdomen and the margins of tergite 5 were slightly concave. With increasing CW, the abdomen of juvenile females progressively became oval and then changed to a rounded shape, and the margins of tergite 5 were clearly convex. Adult females formed a distinct cloud towards the upper right-hand side of the scattergram (Fig. 1A).

Statistical analysis confirmed the curvilinear pattern of the relation between ln(AW) and ln(CW) for juvenile females. A simple regression model and a second-order polynomial regression model did not fit data (lack of fit test, $P < 0.001$ in both cases), whereas a third-order polynomial regression model adjusted to data very well (lack of fit test, $P = 0.988$):

$$\ln(\mathrm{AW}) = -0.100(\ln(\mathrm{CW}))^3 + 0.966(\ln(\mathrm{CW}))^2 - 1.635\ln(\mathrm{CW}) + 0.799,\ N = 1676,\ r^2 = 0.995,\ P < 0.001.$$

As an alternative approach, we adjusted three straight lines to the data set. This was done by applying the method of Somerton and MacIntosh (1983) to all of the data either to the left of the approximate point of positive line curvature or to the right of the approximate point of negative line curvature, along with a subset of data extending from the designated point of curvature to near the other point of curvature. The first and second lines intersected at 15.4 mm CW or 4.5 mm AW, and the second and third lines intersected at 42.5 mm CW or 20.2 mm AW, thereby demarcating three groups of juvenile females designated δ, ε, and ϕ for convenience (Fig. 1). Highly significant regressions of ln(AW) against ln(CW) were obtained for each of the three groups of juvenile females and for the group of adult females (Table 1). The slopes of regressions were greater than one for juvenile females (t test, $P < 0.001$ for each of the three groups); abdomen growth relative to carapace was characterized successively by low positive allometry in the δ group (1.241 ± 0.009; mean slope ± SE), high positive allometry in the ε group (1.488 ± 0.006), and again low positive allometry in the ϕ group (1.262 ± 0.032). However, the slope of the regression of ln(AW) on ln(CW) for adult females did not differ significantly from one, indicating an isometric relationship. The different trends in relative abdomen growth for juvenile females are obvious in Fig. 1B, which also shows that adult females achieved a size-independent mean AW/CW

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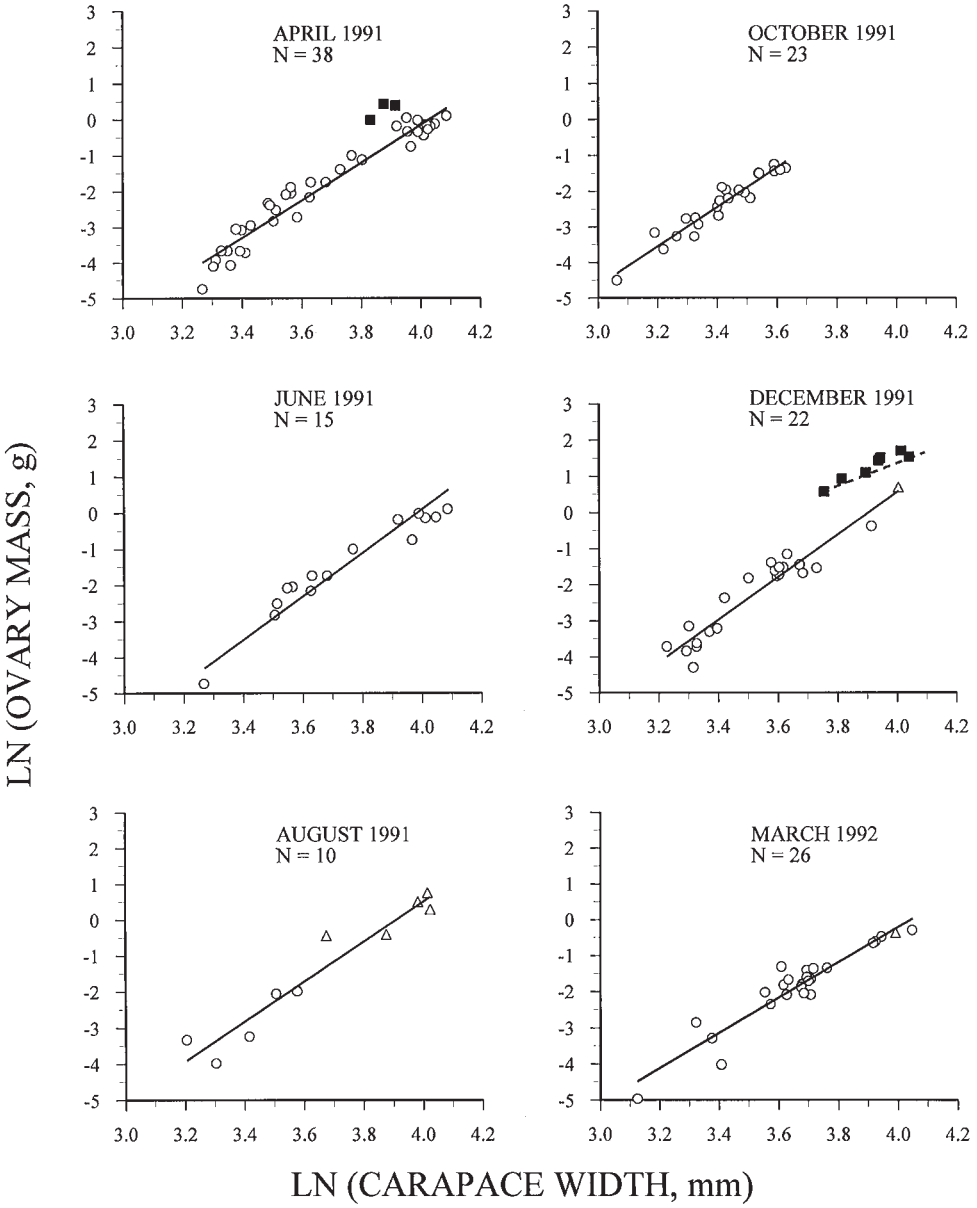
Table 1. Regressions of natural logarithm of abdomen width (AW; in mm) against natural logarithm of carapace width (CW; in mm) for female *Chionoecetes opilio* collected in Baie Sainte-Marguerite from April 1991 to March 1992.

Allometry, maturity, and size categories	Abdomen allometry	r^2	F	N
δ juvenile (CW < 15.4)	ln(AW) = 1.241 ln(CW) – 1.895	0.959	20495.6***	876
ε juvenile (15.4 ≤ CW ≤ 42.5)	ln(AW) = 1.488 ln(CW) – 2.571	0.990	70759.9***	706
ϕ juvenile (CW > 42.5)	ln(AW) = 1.262 ln(CW) – 1.725	0.944	1534.8***	94
Adult	ln(AW) = 1.031 ln(CW) – 0.520	0.910	14232.3***	1410

Note: Slopes of regressions were compared by analysis of variance. r^2, coefficient of determination; F, F statistic; N, sample size. ***, $P < 0.001$.

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Fig. 2. Scattergrams of ln(ovary mass, OM; in g) against ln(carapace width, CW; in mm) for subsamples of female *Chionoecetes opilio* collected in Baie Sainte-Marguerite from April 1991 to March 1992. Ovaries varied in color from white to lilac (○) and from yellow-brown (△) to bright orange (■). Solid lines in graphs represent regressions for females with white to yellow-brown ovaries (see equations and statistics in Table 2). Broken line in December 1991 graph represents the regression line relating ln(OM) to ln(CW) for females immediately after the molt to maturity and prior to oviposition (from Sainte-Marie and Lovrich 1994).



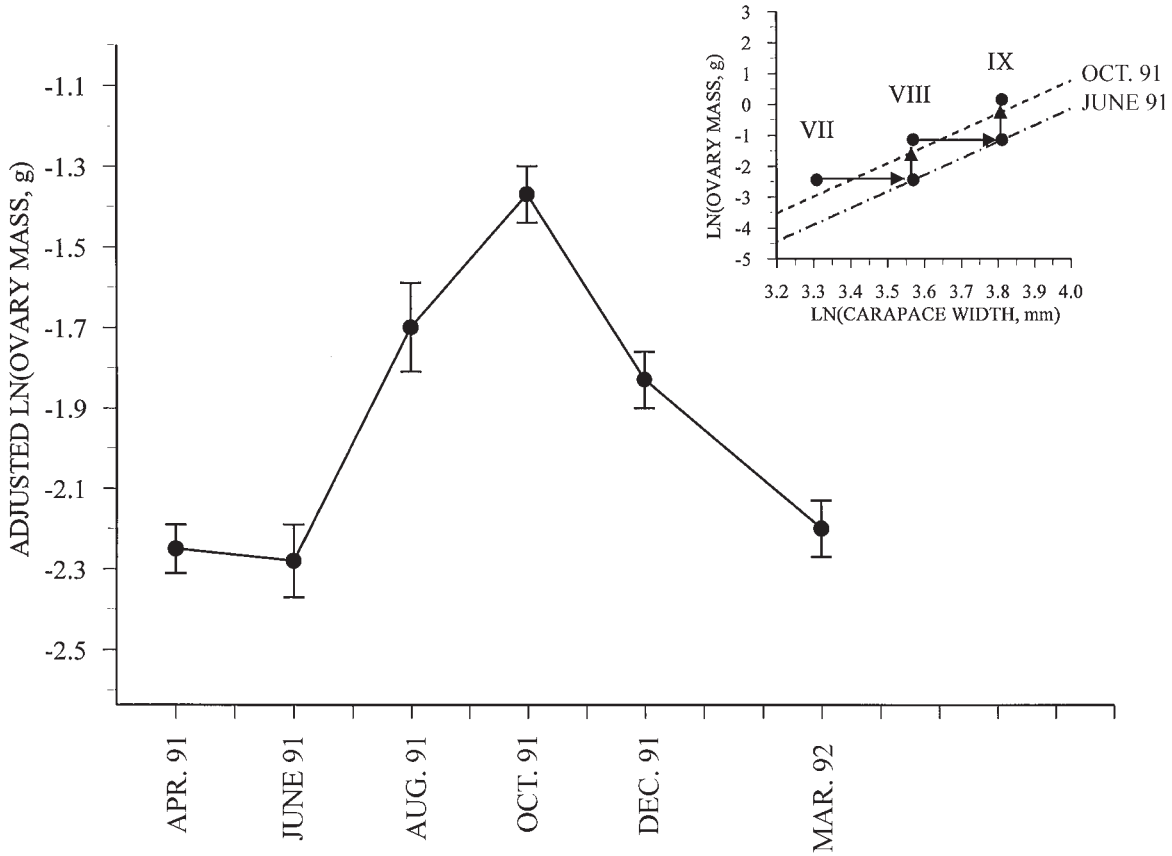
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Table 2. Regressions of natural logarithm of ovary mass (OM; in g) against natural logarithm of carapace width (CW; in mm) for female *Chionoecetes opilio* with rosy-white, lilac, and yellow-brown ovaries collected in Baie Sainte-Marguerite from April 1991 to March 1992.

Date	Regression	r^2	F	N
April 1991	ln(OM) = 5.262 ln(CW) – 21.194	0.949	666.42***	38
June 1991	ln(OM) = 6.028 ln(CW) – 24.017	0.844	70.29***	15
August 1991	ln(OM) = 5.552 ln(CW) – 21.708	0.929	104.66***	10
October 1991	ln(OM) = 5.512 ln(CW) – 21.188	0.922	247.68***	23
December 1991	ln(OM) = 5.880 ln(CW) – 22.970	0.910	203.57***	22
March 1992	ln(OM) = 4.876 ln(CW) – 19.726	0.893	200.98***	26
	Slopes		1.08ns	
	Elevations		1475.63***	

Note: Slopes and elevations were compared by analyses of variance and covariance, respectively. r^2, coefficient of determination; F, F statistic; N, sample size. ns, $P > 0.05$; ***, $P < 0.001$.

Fig. 3. Mean ln(ovary mass; in g) ± SE adjusted to constant carapace width (CW; in mm) by analysis of covariance (see Table 2) for subsamples of female *Chionoecetes opilio* collected in Baie Sainte-Marguerite from April 1991 to March 1992. Insert represents adjusted regression lines for months in which females had smallest (June 1991) or largest (October 1991) ovaries at constant body size. Actual mean CW for instars VII, VIII, and IX from Table 3 are plotted as natural logarithms to show the interaction between carapace and ovary growth. Horizontal arrows represent incremental carapace growth at molt and vertical arrows represent ovary growth between molts.



ratio of 0.683 ± 0.024 (N = 1410). Similar trends were apparent for laboratory females.

Development of ovaries

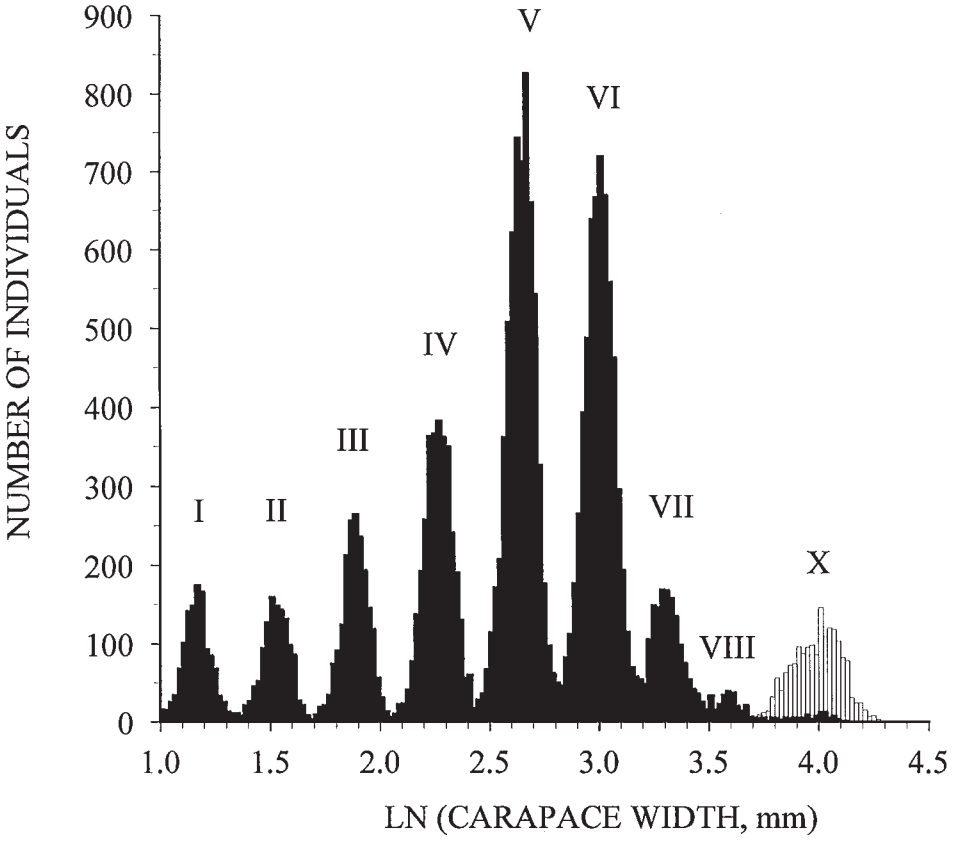
Juvenile females in the 1991–1992 samples from Baie Sainte-Marguerite fell into three size categories on the basis of ovary development. No ovaries were detected by naked eye in females <21.4 mm CW, ovaries were either undetectable or appeared as minute strands in females of 21.4–28.9 mm CW, and ovaries were easily seen in all females >28.9 mm CW.

With increasing size, ovaries changed in color from white to rosy-white, graded into lilac and then into yellow-brown, and finally became bright orange. The largest juvenile female with lilac ovaries was 59.6 mm CW, and the smallest juvenile female with bright orange ovaries was 42.8 mm CW. Only a few juvenile females with bright orange ovaries were collected,

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Fig. 4. Cumulative size-frequency distribution for unsexed and female *Chionoecetes opilio* in Baie Sainte-Marguerite from April 1991 to May 1992. Carapace widths were transformed to their natural logarithm and grouped by logarithmic size-classes of 0.02 units either as unsexed individuals and juvenile females (solid bars) or as adult females (open bars). Note the scarcity of juvenile females in the abscissa range of 3.5–4.2 logarithmic units. Juvenile instars I–VIII and X are identified (N = 23 251).



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in April and December 1991 (Fig. 2). Nevertheless, on these two dates, for a constant CW, ovary size was clearly greater in females with bright orange ovaries than in females with rosy-white to yellow-brown ovaries. Values for ln(OM) in females with bright orange ovaries during December 1991 exceeded mean values measured by Sainte-Marie and Lovrich (1994) in nulliparous females immediately after the puberty molt (Fig. 2). All juvenile females with bright orange ovaries in the December 1991 sample had a second shell and thus were preparing to molt to maturity.

Highly significant regressions of ln(OM) against ln(CW) were obtained on each sampling date for juvenile females with rosy-white to yellow-brown ovaries (Fig. 2, Table 2). From statistical comparison of regression lines, it was apparent that slopes were homogenous but that elevations were heterogeneous over months (Table 2). Adjusted values of ln(OM) plotted against date revealed a general trend of increasing ovary mass from June to October 1991, followed by a decline in ovary mass through December 1991 and March 1992 (Fig. 3). Adjusted ln(OM) did not differ between April and June 1991 (P = 0.82), between April 1991 and March 1992 (P = 0.54), and between June 1991 and March 1992 (P = 0.49), suggesting a 1-year cycle in ovary mass.

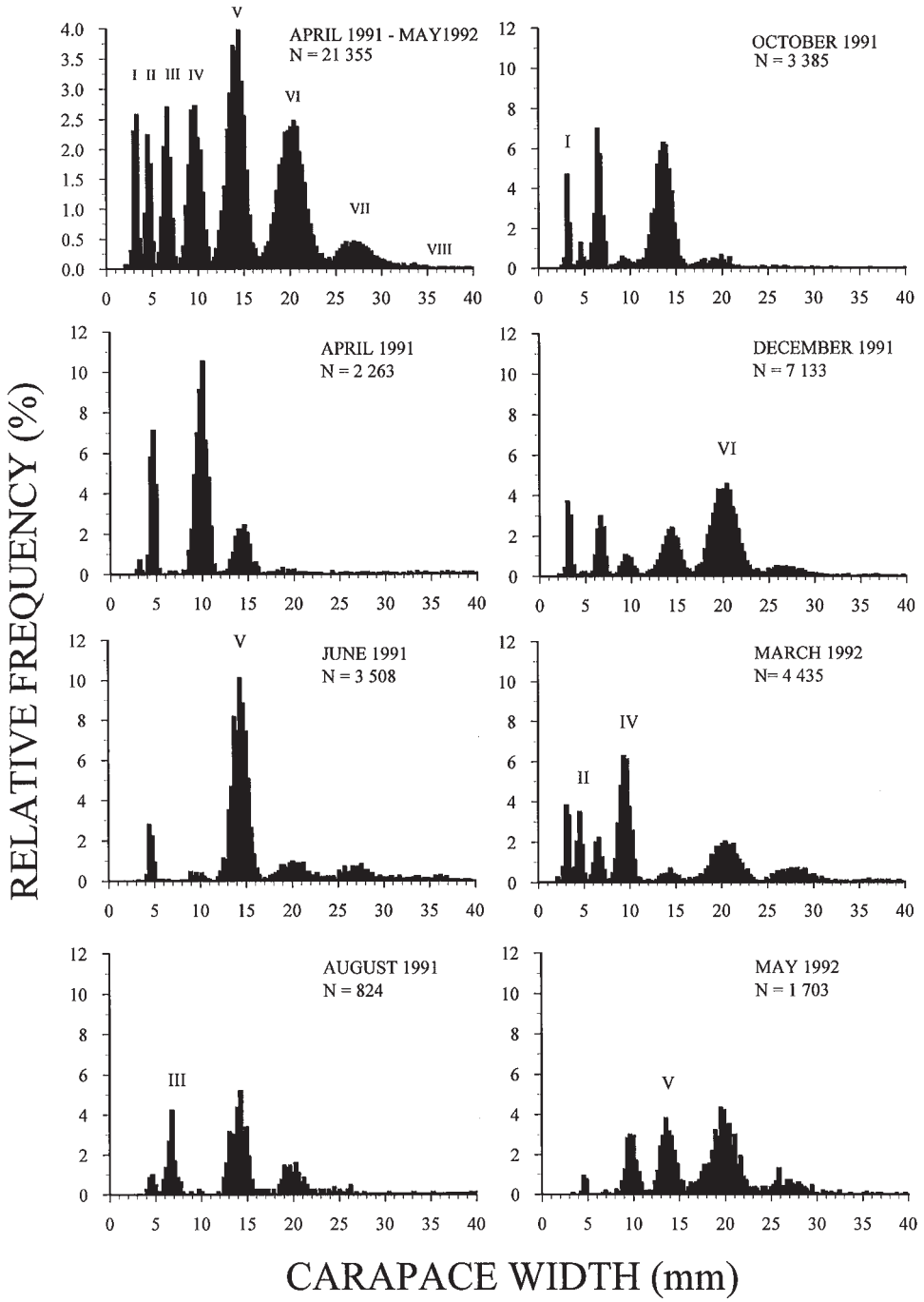
Instars inferred from size-frequency distributions

Carapace width was measured on a total of 23 251 unsexed and female *C. opilio* captured from April 1991 to May 1992 in Baie Sainte-Marguerite (Figs. 4 and 5). More than 91% of females collected were juveniles <32 mm CW. The largest juvenile female measured 62.5 mm CW, but overall, juvenile females >32 mm CW (i.e., 3.47 ln units) were very scarce (Fig. 4). The size-frequency distributions for juvenile females revealed eight outstanding modes, designated instars I–VIII (Fig. 4), while another mode, designated instar X, could be discerned (Fig. 4). Adult females (CW_{min} = 39.6 mm; CW_{max} = 73.5 mm) represented only 7.3% of the total number of captured females. No modes were readily apparent in the size-frequency distribution for adult females (Fig. 4). However, using modal analysis on 1-mm CW size-classes we were able to resolve three modes, designated instars IX–XI (Table 3). The standard deviations about the mean CW for each mode were much higher for instars IX (i.e., 3.37 mm) and X (i.e., 4.17 mm) than for instar XI (i.e., 1.75 mm).

To improve our understanding of the size structure of the largest juvenile females and of adult females, we examined size-frequency distributions for the Baie Sainte-Marguerite population in 1995 and 1996 (Fig. 6). Again, several modes were outstanding for juvenile females. However, compared with 1991–1992, the CW distribution in 1995–1996 was shifted toward larger sizes such that females <16 mm CW were relatively uncommon while those >32 mm CW were well represented. Modal analysis for juvenile females provided

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Fig. 5. Size-frequency distributions by sampling date for unsexed and female *Chionoecetes opilio* <40 mm carapace width in Baie Sainte-Marguerite from April 1991 to May 1992. Instars I–VIII are identified by roman numerals in the upper left summary histogram combining data from all sampling periods. Roman italic numerals above the modes in histograms for individual sampling periods highlight the appearance or marked increase in the relative abundance of a given instar.



statistics for instars IV–IX in 1995 and for instars VI–IX in 1996 (Table 3). The agreement among years between mean CW for each instar was good except for juvenile females belonging to instars VI and IX, which were noticeably smaller in 1996 than in 1991–1992 and 1995 (Figs. 5 and 6, Table 3).

The size structure of primiparous females in 1996 provided direct evidence for two modes at 45.8 and 55.7 mm CW (Fig. 6, Table 3) and corroborated results from the modal analysis for adult females in 1991–1992. It is important to note that, within years, mean CW for primiparous females in instar IX equaled or exceeded those values obtained for juvenile females belonging to the same instar (Table 3). However, modes for primiparous females were jagged compared with those obtained for juvenile females. Closer scrutiny of 1996 data for primiparous females, suggested that both modes might be composite, each comprising two peaks to either side of the

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Table 3. Mean (with SD given in parentheses) of carapace width (in mm) and age at molt measured in months or years for successive instars of postlarval juvenile and (or) adult female *Chionoecetes opilio* in Baie Sainte-Marguerite in 1991–1992, 1995, and 1996.

	1991–1992		1995	1996		
Instar	Juvenile	Adult	Juvenile	Juvenile	Adult	Age
I	3.22 (0.24)	—	—	—	—	0 months (settler)
II	4.63 (0.28)	—	—	—	—	5–7 months
III	6.62 (0.39)	—	—	—	—	12 months
IV	9.67 (0.65)	—	9.18 (1.75)	—	—	17 months
V	14.14 (0.91)	—	14.39 (1.01)	—	—	22 months
VI	20.20 (1.35)	—	20.61 (1.58)	18.14 (1.45)	—	28 months
VII	27.35 (2.01)	—	27.24 (1.87)	27.49 (1.76)	—	3.3 years
VIII	35.50 (2.04)	—	35.69 (2.95)	35.74 (1.27)	—	4.3–4.5 years
IX	45.22 (9.10)	48.46 (3.37)	46.31 (4.35)	44.37 (2.65)	45.85 (3.35)	5.5 years
X	—	57.09 (4.17)	—	—	55.68 (3.22)	6.5 years
XI	—	67.96 (1.75)	—	—	—	7.5 years

Note: Values of CW were derived from modal analyses of size-frequency data shown in Figs. 4 and 6.

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mean CW value. Using modal analysis, we were able to force two submodes to each of the principal modes in the size-frequency distribution for primiparous females: 45.1 ± 3.0, 49.6 ± 1.1, 55.1 ± 1.6, and 59.8 ± 1.1. Adjustment to size-frequency data of the model incorporating four submodes was equally as good ($\chi^2 = 2.80$, $P = 0.9997$) as the model with only two modes ($\chi^2 = 8.75$, $P = 0.9981$).

Molting periods

Over the period 1991–1992, there occurred obvious shifts in the relative importance of the first seven instars in Baie Sainte-Marguerite (Fig. 5). Instar I was not found in August; was scarcely represented in April, May, and June; but was common in October, December, and March. Instar II was relatively abundant from March to June, excepting May, but was uncommon from August to December. Instar III was rare from April to June and well represented from August to March. Instar IV was generally uncommon from June to December but was found frequently from March to April. The relative abundance of instar V was greatest from May to October and then declined from December to April. Instar VI was well represented from December to May but rather uncommon in other months. Instar VII was scarcely represented from April to October 1991 but occurred more frequently in December 1991 and in March and May 1992. The eighth instar was scarce over the whole 1991–1992 sampling period and showed no temporal pattern of abundance. From our bimonthly size-frequency distributions, the following periods of peak molting activity for females were derived: March for *C. opilio* molting from instar I to II and from instar III to IV, May–June for instar IV to V; August for instar II to III, and December for instar V to VI. The time periods for molting of instars VI and VII could not be determined from the size-frequency data.

Direct indicators of molting activity, i.e., the presence in bimonthly samples of molting or soft-shelled females, corroborated the molting schedule inferred above for instars I–V and provided additional information on molting activity of instars VI–VIII (Table 4). Peak molting activity occurred in April for unsexed *C. opilio* passing from instar I to II and for females passing from instar IV to V, in August for unsexed *C. opilio* and females passing from instar II to III, and in December for females passing from instars III to IV, V to VI, VI to VII, VII to VIII, and VIII to IX. However, lesser proportions of juvenile females belonging to instars II–VI molted to the next instar at other times of the year, whereas molting of juvenile females in larger instars appeared to be concentrated mainly in winter. Recently molted adult females were encountered in trawl samples only in December 1991 (0.6% of 164 adult females) and March 1992 (0.8% of 267 adult females). From these molting periods, and by reference to the model developed for males (Sainte-Marie et al. 1995), a post-settlement age can be estimated for successive female instars (Table 3).

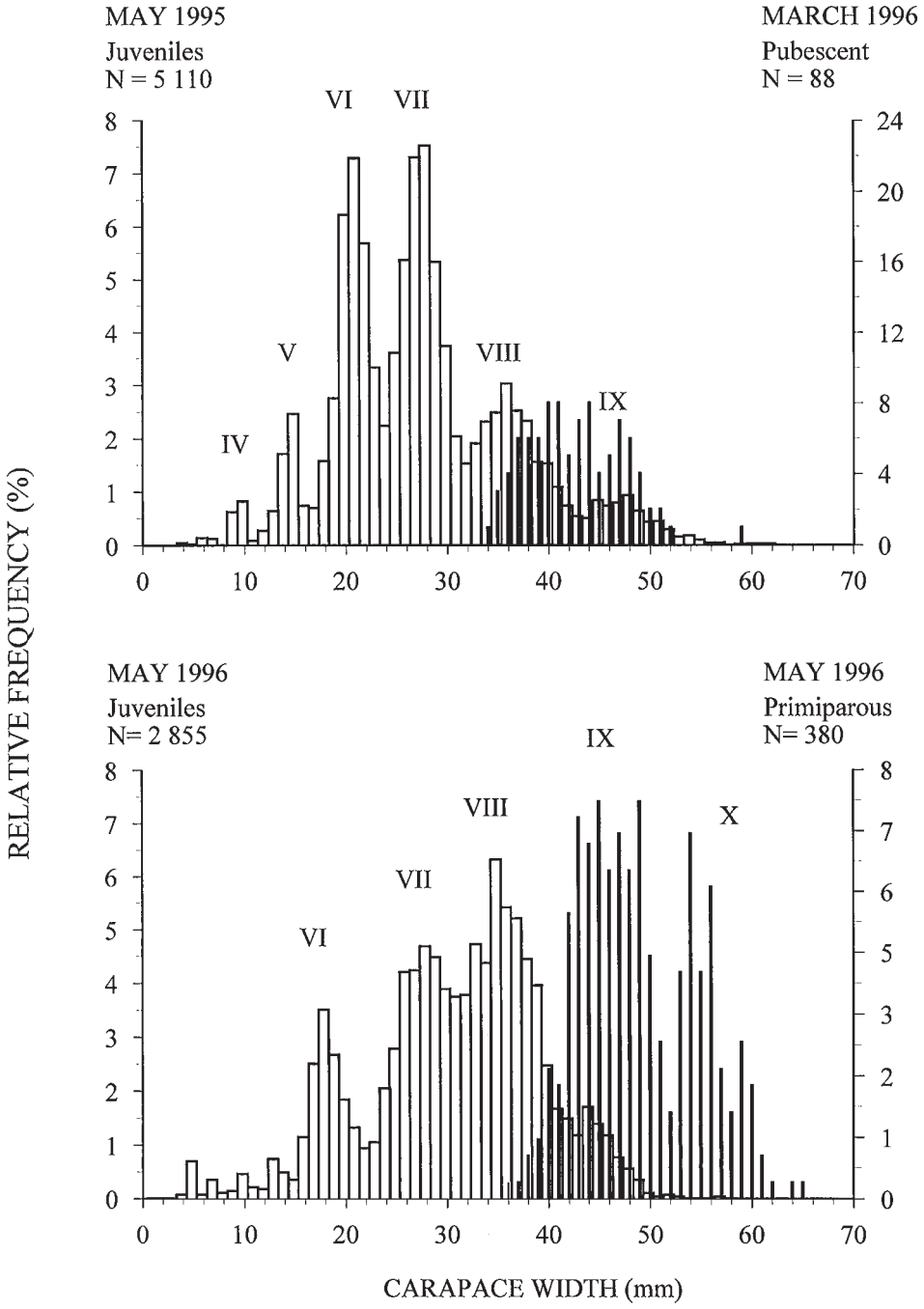
Carapace and abdomen growth per molt

A total of 237 juvenile females of 3.2–67.8 mm premolt CW molted in the laboratory between October 1992 and April 1993 (Figs. 7B and 7C). Of these, 127 molted to a juvenile stage, and the remainder molted to maturity. Moreover, 28 pubescent females of 45.2–62.5 mm premolt CW were collected by SCUBA divers in March 1992 and molted on board the research vessel. Molt increments did not differ significantly between pubescent females from the field and laboratory (Alunno-Bruscia 1993).

Hiatt modeling of carapace growth at molt (Fig. 7B) addressed four groups of laboratory-held females based on ovary development, abdomen allometry, and maturity status after molt: (*i*) immature δ females, (*ii*) prepubescent ε females over the CW range for which all such individuals had detectable ovaries (i.e., 26–42.5 mm CW), (*iii*) prepubescent ϕ females, and (*iv*) pubescent females. Significant Hiatt models were obtained for each of the four groups of females (Table 5). The slopes of Hiatt models differed significantly between immature females and prepubescent ε females, and these two groups of females had significantly steeper slopes than both prepubescent ϕ females and pubescent females. The intersection between the Hiatt models for immature and prepubescent ε females occurred at 16.9 mm CW. Slopes of Hiatt models did not differ significantly between prepubescent ϕ females and pubescent females; however, elevations were significantly different. Average relative carapace molt increments declined progressively in the four groups of females (Fig. 8A): from 54 to 41.4% in immature females over the range 3–15.4 mm premolt CW, from 35.6 to 31.3% in prepubescent females over

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Fig. 6. Size-frequency distributions for *Chionoecetes opilio* sampled in Baie Sainte-Marguerite: beam-trawled juvenile females (open bars) in May 1995 (upper panel) and May 1996 (lower panel), diver-collected pubescent females in March 1996 (solid, narrow bars in upper panel), beam-trawled primiparous females in May 1996 (solid, narrow bars in lower panel). Crabs are grouped by size-classes of 1-mm carapace width and instars are identified in roman numerals above modes.



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the range 26–42.5 mm premolt CW, from 31.3 to 23.2% in prepubescent females over the range 42.5–51 mm premolt CW, and finally from 17.5 to 14.7% in pubescent females over the range 49–67 mm premolt CW.

The same modeling procedure applied to abdomen growth at molt also revealed significant differences among the four groups of females (Table 5, Fig. 7C). On average, relative abdomen molt increments increased from 56 to 66.4% in immature females over the range 0.6–4.5 mm AW (i.e., 3–15.4 mm premolt CW), decreased from 63.9 to 47.6% in prepubescent females over the range 9.7–20.2 mm AW (i.e., 26–42.5 mm premolt CW), and declined even more steeply from 47.6 to 35.6% in prepubescent females over the range 20.2–25.5 mm AW (i.e., 42.5–51 mm premolt CW) (Fig. 8B). However, the relative abdomen molt increment of pubescent females was much larger than that of similarly sized prepubescent females, and declined from 73.4 to 52% over the range 24.2–35.9 mm premolt AW (49–67 mm premolt CW). The differences among

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Table 4. Percent molting activity for unsexed and female *Chionoecetes opilio* in the first eight postlarval instars in Baie Sainte-Marguerite from April 1991 to May 1992.

	Percent molting activity							
Date	I→II	II→III	III→IV	IV→V	V→VI	VI→VII	VII→VIII	VIII→IX
Apr. 1991	4.1	0.0	0.1	6.4	4.2	1.3	0.0	4.3
	(466)	(429)	(1118)	(1469)	(406)	(74)	(59)	(47)
June 1991	1.4	0.9	1.3	0.4	0.2	0.4	0.0	0.0
	(218)	(224)	(77)	(2270)	(2581)	(691)	(394)	(87)
Aug. 1991	0.0	5.1	1.7	0.0	0.3	0.0	0.0	0.0
	(22)	(118)	(114)	(266)	(369)	(124)	(16)	(4)
Oct. 1991	0.0	0.4	0.0	0.0	0.1	0.9	0.0	0.0
	(350)	(797)	(835)	(1687)	(1753)	(215)	(26)	(1)
Dec. 1991	0.0	3.4	10.4	5.3	14.9	8.6	8.9	4.6
	(601)	(763)	(1088)	(1672)	(4787)	(3933)	(485)	(43)
Mar. 1992	1.6	0.1	1.3	1.3	0.7	1.8	0.6	0.0
	(771)	(701)	(1645)	(1509)	(1226)	(1461)	(524)	(102)
May 1992	0.0	0.0	1.2	3.2	0.9	1.2	0.0	0.0
	(31)	(35)	(256)	(678)	(1149)	(879)	(167)	(13)

Note: Molting activity for instars I to VIII is the sum of the number of molting crabs and exuviae in one instar and of the number of soft-shelled crabs in the following instar, expressed as a percentage of total number (in parentheses) of crabs in both instars. Maximum values for molting activity by instar are underlined.

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groups of females in the relative molt increment of either carapace or abdomen, and the differences in relative molt increments between the two body parts for any group of females, are consistent with observed trends in abdomen allometry. In particular, the proportionately greater reduction in relative molt increments of the abdomen compared with the carapace over the combined ε and ϕ prepubescent phases can explain the asymptotic trend in the AW–CW relationship (Fig. 1B).

Mean CWs derived for successive instars from modal analysis of field data were also plotted to form Hiatt growth models (Fig. 7A). The slope of the Hiatt model for immature females:

$$CW_{T+1} = 1.432CW_T + 0.245,\ N = 7,\ r^2 = 0.995,\ P < 0.001$$

was greater than that for combined prepubescent and pubescent females,

$$CW_{T+1} = 1.160CW_T + 4.475,\ N = 11,\ r^2 = 0.990,\ P < 0.001$$

There was no apparent difference between the Hiatt models for prepubescent and pubescent females, suggesting that carapace molt increments in nature were equally as high during the terminal than during regular molts. Given this apparent contradiction with our laboratory observations, we decided to carefully examine the molting patterns for juvenile females in the penultimate and ultimate instars.

The size-frequency distribution for pubescent females collected by SCUBA diving in March 1996 was superimposed on the May 1995 size-frequency distribution for juvenile females (Fig. 6). This was done because females that were pubescent in March 1996 were undoubtedly represented as juvenile (prepubescent) females in the May 1995 size-frequency distribution. From the superimposed CW distributions, it was evident that pubescent females collected in 1996 belonged mostly to instars VIII and IX, with the largest individual (58.9 mm CW) probably belonging to instar X. The fact that the size distribution for instar VIII pubescent females was conspicuously shifted to the right of the size distribution for instar VIII juvenile females suggested that only the largest females of this instar became pubescent in March 1996. This is consistent with the observation that juvenile females belonging to instar IX in May 1996 were exceptionally small (Table 3, Fig. 6) and that primiparous and juvenile females of instar IX were equally as large in May 1996, in spite of lesser molt increments at maturity than at regular molts. By contrast, for instar IX the congruent size distributions of juvenile females in May 1995 and of pubescent females in March 1996 suggests instead that juvenile females in this instar became pubescent irrespective of size. The fact that virtually no juvenile female belonging to instar X was found in May 1996 further suggests that most juvenile females in instar IX molted to maturity in early 1996.

Discussion

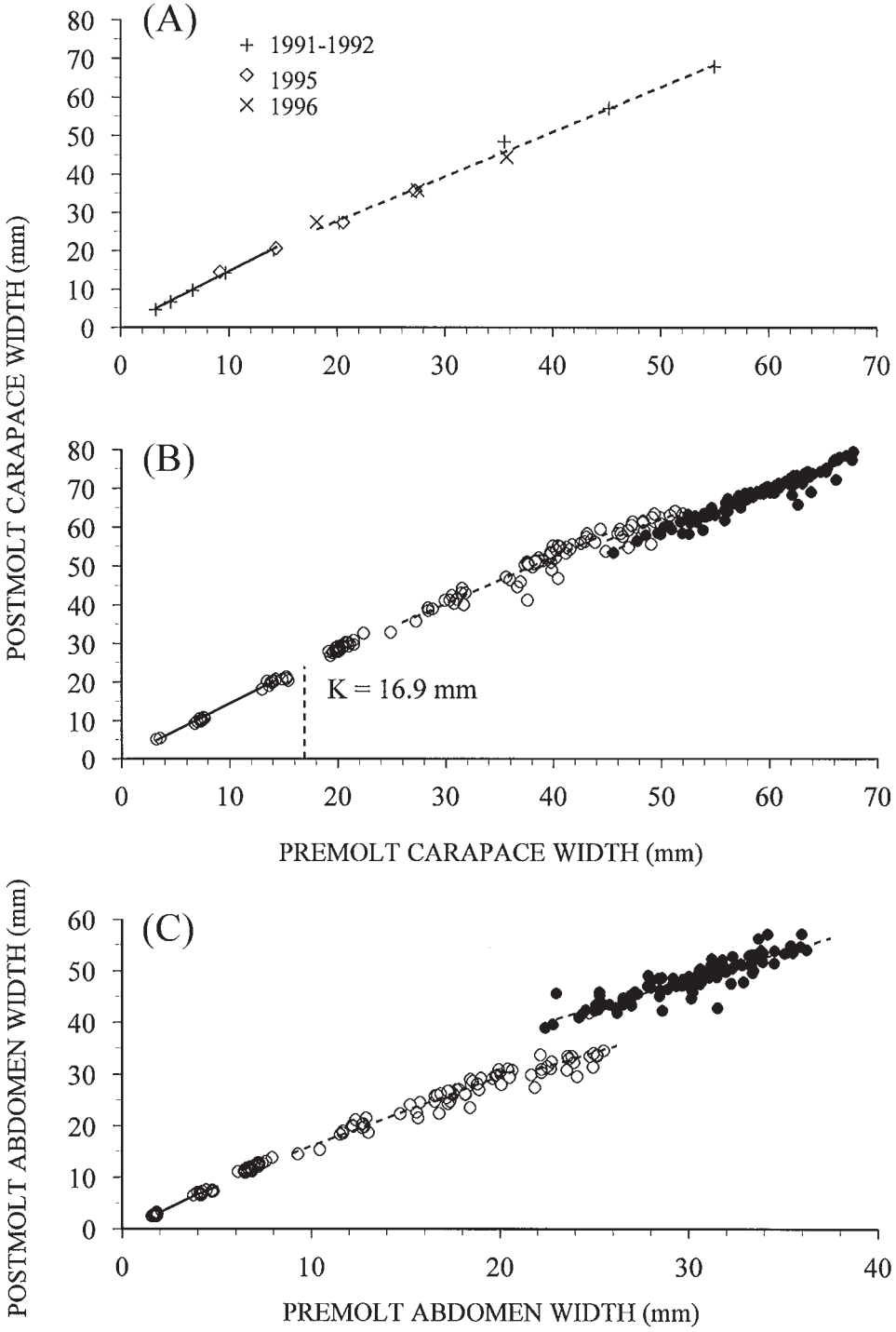
From the extensive data set presented herein emerges a pattern of growth and maturation for female *C. opilio* consisting in three major developmental stages, which we refer to as immature, prepubescent (maturing), and adult. The three developmental stages are typical of majids and have different morphological and physiological attributes, apparent in secondary sexual characters, ovary development and patterns of carapace growth per molt. Below, we discuss the following for females: (*i*) the allometry of abdomen, (*ii*) some aspects of germinal growth, (*iii*) carapace and abdomen growth patterns, and finally (*iv*) we offer a comparison of developmental patterns between females and males.

Abdomen allometry

Basically, the growth of the abdomen relative to carapace in female *C. opilio* follows the general brachyuran pattern and consists in two discontinuous phases with different levels of allometry (Hartnoll 1982). The first phase of positive growth allometry occurs in juvenile females, while a second phase of size isometry is observed for adult females (Fig. 1). The discontinuity between the two phases is caused by a marked

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Fig. 7. *Chionoecetes opilio*. (A) Hiatt carapace growth models for wild-caught immature (solid line) and prepubescent and pubescent females (broken line) using successive estimates of size at instar from Table 3. The middle (B) and lower (C) panels represent carapace and abdomen growth per molt measured on females held in the laboratory from October 1992 to April 1993. In the two lower panels, open circles indicate a regular molt, solid circles indicate a molt to maturity, while the oblique solid and the two first oblique broken lines starting from the left represent the Hiatt growth models for immature and ε and ϕ prepubescent females, respectively (see Table 5). The vertical broken line in the middle pane gives the abscissa value for the intersection of the regressions for immature and ε prepubescent females.



increase in abdomen size to accommodate the egg clutch, which occurs at the molt to maturity (Figs. 7C and 8B; Watson 1970; Moriyasu et al. 1987).

The pattern of growth allometry of the abdomen of juvenile females can further be divided into three subtle but nevertheless biologically meaningful phases (Fig. 1) by a positive inflection at 15.4 mm CW (i.e., 4.5 mm AW) and a negative inflection at 42.5 mm CW (i.e., 20.2 mm AW). These two

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Table 5. Hiatt growth models for immature, prepubescent and pubescent female *Chionoecetes opilio* held in the laboratory from October 1992 to April 1993.

Reference number and maturity and allometry categories	Carapace growth	Abdomen growth
(1) Immature, δ	$CW_{T+1} = 1.429CW_T - 0.227$ $r^2 = 0.994$, $F = 3932.4$***, $N = 27$ $CW < 15.4$	$AW_{T+1} = 1.728\ AW_T - 0.287$ $r^2 = 0.989$, $F = 1838.1$***, $N = 23$ $AW < 4.5$
(2) Prepubescent, ε	$CW_{T+1} = 1.246CW_T + 2.864$ $r^2 = 0.942$, $F = 697.3$***, $N = 45$ $26 < CW < 42.5$	$AW_{T+1} = 1.326AW_T + 3.033$ $r^2 = 0.931$, $F = 634.9$***, $N = 49$ $9.7 < AW < 20.2$
(3) Prepubescent, ϕ	$CW_{T+1} = 0.828CW_T + 20.585$ $r^2 = 0.754$, $F = 85.8$***, $N = 30$ $CW \geq 42.5$	$AW_{T+1} = 0.944AW_T + 10.496$ $r^2 = 0.366$, $F = 13.3$**, $N = 25$ $AW \geq 20.2$
(4) Pubescent	$CW_{T+1} = 1.071CW_T + 5.099$ $r^2 = 0.993$, $F = 1511.7$***, $N = 110$	$AW_{T+1} = 1.076AW_T + 15.9430$ $r^2 = 0.809$, $F = 457.9$***, $N = 110$
Slopes	F=14.40*** 1≠2≠(3=4)	F=4.60** 1≠2≠(3=4)

Note: Minimum and (or) maximum values of carapace width (CW; in mm) and of abdomen width (AW; in mm) used for models are given where appropriate. Slopes of regression models were compared by analysis of covariance. CW_{T+1} and AW_{T+1}, postmolt carapace and abdomen widths, respectively; CW_T and AW_T, premolt carapace and abdomen widths, respectively; r^2, coefficient of determination; F, F statistic; N, sample size; **, $P < 0.01$; ***, $P < 0.001$.

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inflections are not artifacts caused by measurement error or atypical data. The first inflection is also clearly seen in laboratory-reared female *C. opilio* that were measured using very precise imaging techniques, while the second inflection is seen consistently in each year or different site for which we have numerous measurements of AW and CW (B. Sainte-Marie, unpublished data). The first inflection marks the start of a period of high positive growth allometry and coincides approximately with the modal size of instar V, which immediately precedes the smallest instar (VI, see below) in which some individuals had ovaries detected by eye. The second inflection, however, marks a decrease in relative abdomen growth and apparently occurs only in females that surpass 42.5 mm CW as juveniles. This downscaling may reflect structural limits on abdomen size. Indeed, extrapolation of the allometric growth function for juvenile ε females would result in abdomens subequal or equal in width to those of adult females at ca. 80 mm CW, a body size that some prepubescent female *C. opilio* may reach in eastern Canada (B. Sainte-Marie, unpublished data). Excessively large abdomens may hinder locomotion (Hartnoll 1982; Gherardi and Micheli 1989), and there is an optimal relative size the abdomen must reach to function effectively with the sternum as an incubatory chamber (Hartnoll 1974). That relative limit for *C. opilio* appears to be an AW/CW ratio of about 0.68 (Fig. 1B). Hence, in a species with females having a potential to become mature over a very broad size range, the ε phase ensures that the abdomen rapidly is scaled up to a size that is within one molt increment's reach of target AW/CW ratio, even for the smallest of pubescent females, while the ϕ phase ensures that the biggest of pubescent females do not become endowed with unnecessarily large abdomens.

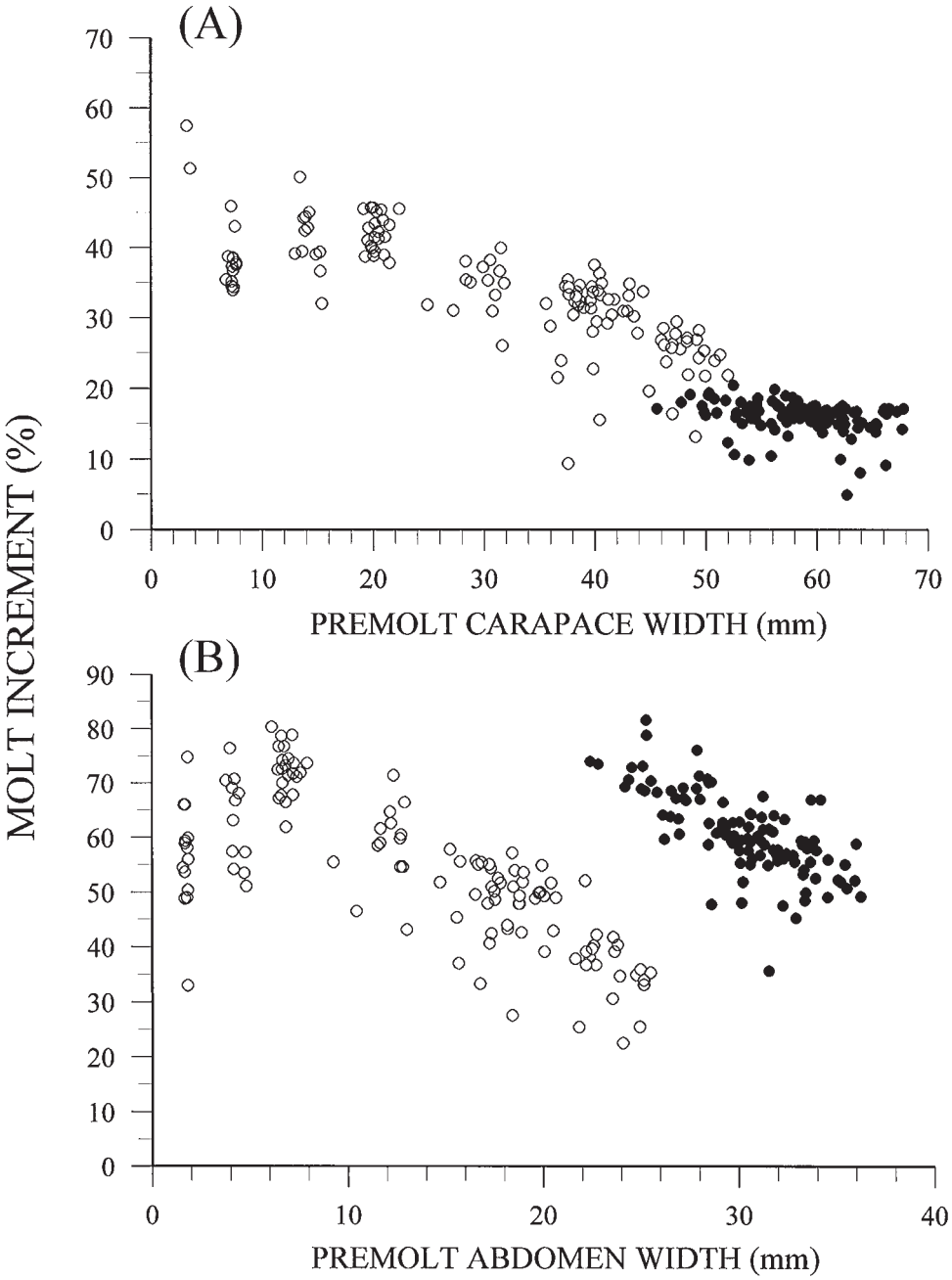
In *C. opilio*, the δ phase in growth allometry coincides roughly with the immature developmental stage, while together the ε and ϕ phases coincide with the prepubescent developmental stage. These phases in growth allometry of abdomen were not detected in previous studies on *C. opilio*, and to our knowledge the ϕ phase or its equivalent has never before been reported for another brachyuran. Ito (1957) recognized immature and prepubescent developmental stages for *C. opilio* based on curvature in the plot of AW against CW, but as Conan et al. (1990) noted, data were not logarithmically transformed such that the curvilinear pattern may result from a simple positive allometric relation that cannot be interpreted in terms of different growth stages. Another study of abdomen allometry for *C. opilio*, although correct in approach, was based only on a few tens of data points over a limited range of body sizes (Watson 1970) compared with more than a thousand data points over the full range of available body sizes in the present study (Fig. 1). Somerton and MacIntosh (1983) and Clayton (1990) pointed out that detection of subtle changes in allometry may require numerous data with an appropriate scope. Only two allometric growth phases are recognized for juvenile females in brachyuran species other than *C. opilio*, and invariably, the first is characterized by low positive allometry and the second by high positive allometry of either chelae or abdomen (e.g., Hartnoll 1963; Pohle and Telford 1982; Haefner 1985; Gaertner and Laloé 1986), as are respectively the δ and ε phases in *C. opilio*. Also, note that the link between allometric and physiological change has not often been demonstrated. However, Haefner (1990) found a reasonably good correlation between changes in abdomen allometry and the onset of ovary development in juvenile females of the portunid crab, *Callinectes ornatus*, although as may be the case for *C. opilio* the change from low to high positive growth allometry apparently preceded initiation of ovary development.

Ovary development and growth

The rather large size range (21.4–28.9 mm CW) over which ovaries were initially detected by naked eye suggests that gonad development started in either of two successive instars. Indeed, based on size-frequency distributions (Fig. 5), females at the lower end of the size range probably represented large individuals of instar VI, while those at the upper end of the size range probably represented small to medium individuals of instar VII. After the ovary became visible, two successive

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Fig. 8. Carapace (A) and abdomen (B) molt increments expressed as a percentage of premolt carapace width for *Chionoecetes opilio* held in the laboratory from October 1992 to April 1993 (same data as in Figs. 7B and 7C). Open circles indicate a regular molt, and solid circles indicate a molt to maturity.



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steps in gonadal development could be recognized. The first is previtellogenesis represented by rosy-white to lilac ovaries, and the second is vitellogenesis represented by yellow-brown to bright orange ovaries. The change in color from lilac to orange reflects oocyte loading with yolk, and is caused by the presence of carotenoids linked to vitellogenin (Meusy and Payen 1988).

The progression in the size of ovaries was more or less continuous through successive prepubescent instars. Indeed, the "cycle" observed in ovary mass over 1 year (Fig. 3) is deceptive and, in fact, reflects alternating periods of germinal growth and incremental body growth. The ascending portion of the "cycle" in adjusted ln(OM), from June through October 1991, coincided with the intermolt period for most prepubescent females. Conversely, adjusted ln(OM) declined markedly in December 1991 and in March 1992, when most prepubescent and pubescent females molted. Thus, the apparent cycle in ovary size emerges because molting causes a sudden increase in body size that is not accompanied by incremental growth of the ovary, as conceptualized in the schematic model of Fig. 3. Ovary mass did not reach the highest values predicted by our model at a given body size, probably because ovary development proceeded for a month or so past our October sampling date before juvenile females began to molt. Our

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model also can explain why pubescent females in December 1991, which were preparing to molt to maturity, had relatively larger ovaries than those observed in recently molted nulliparous females (Fig. 2).

The time period or number of molt cycles required for full ovary development in juvenile females probably varies as a function of definitive body size. Since ovaries began to develop perceptibly in the sixth or seventh instar, and the vast majority of females achieved maturity in the ninth or tenth instar, at least three intermolt periods (i.e., 3 years) were usually required for full ovary development in juvenile females. This is the same number of intermolt periods that Donaldson et al. (1981) thought was necessary for ovaries to develop fully in *Chionoecetes bairdi*. However, the smallest (39.6 mm CW) and largest (73.5 mm CW) adult females represented in the *C. opilio* population of Baie Sainte-Marguerite in 1991–1992 were mature in the 8th or 11th postsettlement molt, respectively. This suggests that full ovary development may occur in as little as 2 years or in as long as 5 years, assuming the size threshold for the onset of ovary development does not vary. In any case, our results are at odds with Moriyasu et al.'s (1987) statement that ovary development in juvenile female *C. opilio* occurs within a single (annual) molt cycle. Full ovary development may indeed occur over a period of 1 year (Mallet et al. 1993; Sainte-Marie and Carrière 1995; Lanteigne et al. 1996), but only in (terminally molted) adult females in which there is no competition for energy between somatic and germinal growth and only provided ambient temperature is in excess of ≈0–1°C.

Growth and age patterns in relation to sexual maturity

Growth and reproduction in Brachyura are antagonistic processes (e.g., Kurata 1962; Hartnoll 1985), and this was clearly reflected in growth patterns of female *C. opilio*. There occurred a significant decrease in the slope of the carapace growth function at the onset of ovary development at 16.9 mm CW, which is halfway between the modal sizes for instars V and VI, and an even more pronounced decline of incremental carapace growth at the molt to maturity (Figs. 7B and 8A). The reduction in carapace growth at onset of ovary development betrays diversion of energy from somatic growth into germinal growth, as discussed for several other brachyurans (e.g., Hartnoll 1982; Gerhardi and Micheli 1989; Wolff and Soto 1992). The much lesser carapace growth increment at the molt to maturity may result from energy investment in abdomen growth and in vitellogenesis, the latter occurring principally during the ultimate intermolt period.

The ϕ phase in abdomen allometry was reflected in growth by a reduction of the molt increment, most obvious in the abdomen but also seen in the carapace. If our biological interpretation of the ϕ phase is correct, i.e., reduction of relative growth to meet structural limits on abdomen size, then it is unclear why there did not occur only a reduction in the growth of the abdomen. Perhaps carapace and abdomen growth cannot be completely uncoupled, and the reduction in carapace molt increment is an "inescapable by-product" (Hartnoll 1982) of the downscaling in relative abdomen growth.

Some uncertainty as to the terminal nature of the maturity molt in female *C. opilio* persists in the literature, because the size differential between the largest juvenile and adult females is perceived to be too great to be accounted for by one molt increment (Elner and Beninger 1995) or because evidence in support of the anecdysic condition of adult females is limited (Conan et al. 1990). Also, there exists two reports of molting by adult females, the one considering this to be an atypical event (Ito and Kobayashi 1967) while the other raises doubts about the terminal nature of the maturity molt (Hooper 1986). However, in study sites with which we are extensively familiar, the largest adult females always are within one molt increment's reach of the largest juvenile females, but both types do not necessarily co-occur temporally owing to discontinuous recruitment. Further, dissection throughout the year of several hundred wild-caught adult females with various exoskeleton conditions has never revealed any female with a double shell indicating a potential for molting (Sainte-Marie 1993 and unpublished data). Similarly, adult females kept over several years in our tank facilities have never molted, unlike juvenile females. Finally, the sudden disappearance from populations of adult females 5–6 years following massive recruitment events is consistent with the terminal nature of the maturity molt and provides an estimate for life expectancy after maturity (Somerton 1981*a*; Sainte-Marie et al. 1996). Thus, overall, the evidence in favor of the terminal nature of the maturity molt for females is good.

Modal analysis of the size structure of adult females was complicated by the fact that individual modes were not well differentiated, in stark contrast to the case for juvenile females (Fig. 4). This could result from three nonexclusive processes: (*i*) increasing modal variance with age or maturity, (*ii*) reduction of size differential between successive adult instars owing to a size-dependent terminal molt process, and (*iii*) intercohort variability in size at molt. First, divergent growth trajectories for individuals in a cohort might arise as a result of differences in the size at onset of ovary development (i.e., large individuals of instar VI or small individuals of instar VII). However, seemingly, this would cause variance for individual modes to increase only if it was the smaller females of an instar that initiated ovary development first. This hypothetical pattern is inconsistent with our observation that it is the largest females in a cohort (or instar) that first develop ovaries and undergo terminal molt to maturity (Fig. 6). Conversely, spreading of distributions could also arise if large females are energetically more efficient than small females at instar. For example, in the grapsid crab, *Pachygrapsus marmoratus*, at any given instar, large individuals have relatively greater molt increments than small individuals (Vernet-Cornubert 1958*b*). Second, the propensity for larger females to mature earlier would cause the size differential between adult instars to collapse. Indeed, at a given instar the smaller size immediately prior to molt of late compared with early maturing females would offset the size benefit of a larger molt increment at a regular than at a terminal molt. This phenomenon is clearly seen in our data, which show that juvenile females belonging to instar IX were on average slightly smaller than adult females in the same instar both in 1991–1992 and in 1996 (Table 3). Consequently, the mean size of successive adult instars is progressively less than would be projected from our Hiatt models assuming the size distribution of females molting to maturity through successive prepubescent instars was not skewed. Third and finally, the blurring of modes could result from accumulation of adult (terminally molted) females from several different year-classes with different sizes at a given instar. We detected significant interannual

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Table 6. Comparison of growth and maturation patterns of female and male *Chionoecetes opilio* in Baie Sainte-Marguerite in 1991–1992.

Feature	Female	Male
Developmental stages	3 (immature, prepubescent, adult)	3 (immature, adolescent, adult)
Molt leading to sexual maturity	Terminal molt	Pubertal, preterminal molt
Size for sexual maturity (mm)	40–74[a]	37–39[a]
Size after terminal molt (mm)	40–74[a]	41–135[a]
Most common:		
Instars after terminal molt	IX and X	IX, X, XI, XII, and XIII
Sizes after terminal molt (mm)	45–48 and 56–57[b]	51, 65, 80, 97 and 115[b]
Ages at terminal molt (years)	5.5–6.5	5.7–9.7

Note: Sizes are carapace widths rounded to nearest integer. Data for males are from Sainte-Marie et al. (1995).
[a]Values are the range of the distribution.
[b]Values are modal sizes for each instar.

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variation in size for instar VI (Table 3), which may result from differences in size at settlement that carry through to larger sizes and (or) from differences in carapace growth per molt among different cohorts.

The observation that larger female *C. opilio* in a cohort mature earlier suggests that there may be phenotypic and (or) genotypic constraints on size and age at maturity. For example, in the cladoceran *Daphnia magna*, which also exhibits a terminal molt to maturity, the potential for early or late maturity in females can be traced to maternal effects (Ebert 1994). For *C. opilio*, elucidation of the causes of heterogeneity in patterns of growth and maturation within and among cohorts is underway by examination of growth of individual cohorts in the time series for Baie Sainte-Marguerite and of individual crabs hatched and reared in controlled laboratory conditions.

This study and previous analyses (Alunno-Bruscia 1993; Sainte-Marie et al. 1996) have established that most female *C. opilio* in the northern Gulf of St. Lawrence, over the period 1991–1996, reached maturity at instars IX or X at the postsettlement age of 5.5 or 6.5 years, respectively (Table 3). The size of female *C. opilio* in Baie Sainte-Marguerite at the successive juvenile instars coincides with values derived for this species in two seas of the North Pacific, up until instar VIII. Thereafter, incremental carapace growth of females in the Sea of Okhotsk remained high such that juvenile females in instar IX and X reached mean sizes of 49.5 and 65.5 mm CW, respectively, and matured into instar XI at 75.5 mm mean CW and into instar XII at 84.5 mm mean CW (Ito 1970). In the Sea of Japan, the growth pattern was almost identical to that in the Sea of Okhotsk except for the fact that all adult females apparently belonged to instar XI and had a mean size of 77.4 mm CW (Kon 1980). Estimates of age at instar in both of the preceding studies were noticeably higher than our own, but both studies lacked detailed information on molting schedules.

Several investigators have remarked that female *C. opilio* tend to be smaller at maturity towards the northern limit of their distribution and have thus posited a direct link between temperature and female size at maturity (Jewett 1981; Somerton 1981*a*, 1981*b*; Paul et al. 1997). This hypothesis might explain some measure of temporal and geographical variability for average size at maturity within the Gulf of St. Lawrence system. With abnormally low water temperatures prevailing since 1986, and especially since 1990 (Gilbert and Pettigrew 1997), adult females recruiting to populations in the Estuary and the northern Gulf of the St. Lawrence have in recent years measured between 35 and 65 mm CW (Fig. 6; Sainte-Marie et al. 1996 and unpublished data). These small values for size at maturity stand in contrast to values of 55–85 mm CW in the same general areas from the early 1960s into the early 1980s (e.g., Brunel 1962; Powles 1968; Watson 1970; Haynes et al. 1976). Moreover, in the Saguenay Fjord the mean 76 mm CW size of adult females in 1989 (Sainte-Marie et al. 1992) exceeded the size of the largest adult female found in Baie Sainte-Marguerite over the whole period 1988–1996. The deep upstream basin of the Saguenay Fjord where *C. opilio* occurs has only limited saltwater exchanges with the neighboring St. Lawrence system (Schafer et al. 1990), and water temperature on bottoms inhabited by adult females has averaged about 2°C warmer in the Fjord than in the St. Lawrence Estuary and northern Gulf over the past 10 years (B. Sainte-Marie, unpublished data). Historical and geographical patterns similar to those of females also exist for males (Sainte-Marie et al. 1995). Even though *C. opilio* populations may experience only very small amplitude temperature variations across years, owing to their stenothermic cold-water habit, it is tempting to speculate that these variations are sufficient to induce profound changes in growth and maturation schedules for both females and males.

Comparison of maturation and growth between sexes

Table 6 summarizes some important features of growth and maturation for female and male *C. opilio* in Baie Sainte-Marguerite, based on data from 1991 to 1992. One main difference between the sexes is that functional maturity precedes terminal molt in males, whereas it is strictly dependent on terminal molt in females. Indeed, provided that they are hard shelled (i.e., inter- or pre-molt), adolescent males as small as 40–45 mm CW can copulate successfully with females (Sainte-Marie and Lovrich 1994; B. Sainte-Marie, unpublished data). However, expression of reproductive potential for adolescent males is contingent on individual (e.g., molt status) and social (e.g., male competition and female acceptance) contexts, such that under normal circumstances the smaller adolescent males in a population may rarely participate in reproduction (Ennis et al. 1990; Elner and Beninger 1995). Nevertheless, female and male *C. opilio* may reach functional maturity almost simultaneously, as soon as instar VIII, i.e., 4.5 years after settlement.

Mean CW and age at instar are similar between the sexes up to the eighth instar. However, females entering the ϕ phase

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of abdomen allometry and those undergoing terminal molt have smaller carapace molt increments than similarly sized males, hence the difference in mean CW between females and males belonging to instars IX and X. Male carapace molt increments are not perceptibly less at terminal than at regular molts for a given body size (Hoenig et al. 1994; Sainte-Marie et al. 1995). The marked reduction in carapace molt increments at terminal compared with regular molts for females probably is related to greater investment into germinal growth (Hartnoll 1985). For example, in *C. opilio* of 57 mm CW the maximum mass of gonads for adult males is ≈2.8 g (Sainte-Marie et al. 1995, Fig. 10), compared with mean values of ≈4.4 g in females prior to first oviposition (Sainte-Marie and Lovrich 1994, Fig. 2) and ≈7.9 g in females prior to second or third oviposition (Sainte-Marie 1993, Fig. 3).

Over the period 1988–1996, adult females in Baie Sainte-Marguerite belonged essentially to instars IX and X, whereas adult males were more evenly distributed among at least five successive instars (i.e., IX–XIII; Sainte-Marie et al. 1995, 1996). Given the discontinuous nature of recruitment processes and the existence of terminal molt for *C. opilio,* gender-related differences in size (age) at maturity cause marked interannual variability in the operational sex ratio, intensity of sexual competition, and egg production, which are partly or completely discussed elsewhere (Elner and Beninger 1995; Sainte-Marie and Dufour 1995; Sainte-Marie et al. 1996).

In conclusion, this study has established the basic pattern for maturation and growth of females and some of the most notable differences in growth and maturation between sexes in *C. opilio*. Examination of interannual trends in size progression of female age groups has revealed unsuspected complexities in the maturation and growth patterns, including size-dependent terminal molting. Detection of this last feature was possible largely because females preparing for the terminal molt can be distinguished from those preparing for regular molts, which is unlike the situation in males. Our analysis also shows that single-year or pooled multiyear size-frequency distributions may lead to incomplete or erroneous interpretations of growth: there may be significant and biologically meaningful differences in growth among cohorts that can be interpreted only through cohort analysis and direct measurements of growth per molt. Thus, a more detailed analysis of cohort dynamics may still reveal fundamental aspects of growth and recruitment for both female and male *C. opilio* that have remained unnoticed to date.

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References

Aldrich, J. 1974. Allometric studies on energy relationships in the spider crab *Libinia emarginata* (Leach). Biol. Bull. (Woods Hole), **147**: 257–273.

Alunno-Bruscia, M. 1993. Croissance et maturation des femelles du crabe des neiges, *Chionoecetes opilio* (O. Fabricius), dans le golfe du Saint-Laurent. DEA (Diplôme d'études approfondies) thesis, Université d'Aix-Marseille II, Marseille, France.

Bailey, R.F.J., and Elner, R.W. 1989. Northwest Atlantic snow crab fisheries: lessons in research and management. *In* Marine invertebrate fisheries: their assessment and management. *Edited by* J.F. Caddy. John Wiley & Sons, New York. pp. 261–280.

Bennett, D.B. 1973. The effect of limb loss and regeneration on the growth of the edible crab, *Cancer pagurus* L. J. Exp. Mar. Biol. Ecol. **13**: 45–53.

Bouchard, S., Sainte-Marie, B., and McNeil, J.N. 1996. Indirect evidence indicates female semiochemicals release male precopulatory behaviour in the snow crab, *Chionoecetes opilio* (Brachyura: Majidae). Chemoecology, **7**: 39–44.

Brunel, P. 1962. Troisième série d'observations sur la biologie et la biométrie du crabe-araignée *Chionoecetes opilio* (Fabr.). Rapp. Annu. 1962. Station de biologie marine, Grande-Rivière, Qué. pp. 91–100.

Clayton, D.A. 1990. Crustacean allometric growth: a case for caution. Crustaceana (Leiden), **58**: 270–290.

Conan, G.Y., Elner, R.W., and Moriyasu, M. 1990. Review of the literature on life histories in the genus *Chionoecetes* in light of recent findings on growth and maturity of *C. opilio* in eastern Canada. *In* Proceedings of the International Symposium on King and Tanner Crabs. Lowell Wakefield Fish. Symp. Ser. Alsk. Sea Grant Rep. No. 90-04. University of Alaska, Fairbanks. pp. 163–179.

Donaldson, W.E., Cooney, R.T, and Hilsinger, J.R. 1981. Growth, age and size at maturity of Tanner crab, *Chionoecetes bairdi* M.J. Rathurn, in the northern Gulf of Alaska (Decapoda, Brachyura). Crustaceana (Leiden), **40**: 286–302.

Draper, N.R., and Smith, H. 1981. Applied regression analysis. 2nd ed. John Wiley & Sons, New York.

Ebert, D. 1994. A maturation size threshold and phenotypic plasticity of age and size at maturity in *Daphnia magna*. Oikos, **69**: 309–317.

Elner, R.W., and Beninger, P.G. 1992. The reproductive biology of snow crab, *Chionoecetes opilio*: a synthesis of recent contributions. Am. Zool. **32**: 524–533.

Elner, R.W., and Beninger, P.G. 1995. Multiple reproductive strategies in snow crab, *Chionoecetes opilio*: physiological pathways and behavioral plasticity. J. Exp. Mar. Biol. Ecol. **193**: 93–112.

Ennis, G.P., Hooper, R.G., and Taylor, D.M. 1990. Changes in the composition of snow crab (*Chionoecetes opilio*) participating in the annual breeding migration in Bonne Bay, Newfoundland. Can. J. Fish. Aquat. Sci. **47**: 2242–2249.

Food and Agriculture Organization of the United Nations (FAO). 1993. Fishery statistics—catches and landings. Vol. 76. FAO, Rome. pp. 1–687.

Gaertner, D., and Laloé, F. 1986. Étude biométrique de la taille à première maturité sexuelle de *Geryon maritae* Manning et Holthuis, 1981 du Sénégal. Oceanol. Acta, **9**: 479–487.

Gherardi, F., and Micheli, F. 1989. Relative growth and population structure of the freshwater crab, *Potamon potamios palestinensis*, in the Dead Sea area (Israel). Isr. J. Zool. **36**: 133–145.

Gilbert, D., and Pettigrew, B. 1997. Interannual variability (1948–1994) of the CIL core temperature in the Gulf of St. Lawrence. Can. J. Fish. Aquat. Sci. **54**(Suppl. 1): 57–67.

Haefner, P.A., Jr. 1985. Morphometry, reproduction, diet, and epizoites of *Ovalipes stephensoni* Williams, 1976 (Decapoda, Brachyura). J. Crustacean Biol. **5**: 658–672.

Haefner, P.A., Jr. 1990. Morphometry and size at maturity of *Callinectes ornatus* (Brachyura, Portunidae) in Bermuda. Bull. Mar. Sci. **46**: 274–286.

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Hartnoll, R.G. 1963. The biology of Manx spider crabs. Proc. Zool. Soc. London, **141**: 423–496.

Hartnoll, R.G. 1974. Variation in growth pattern between some secondary sexual characters in crabs (Decapoda, Brachyura). Crustaceana (Leiden), **27**: 131–136.

Hartnoll, R.G. 1982. Growth. *In* The biology of Crustacea. Vol. 2. *Edited by* L.G. Abele. Academic Press, New York. pp. 11–96.

Hartnoll, R.G. 1985. Growth, sexual maturity and reproductive output. *In* Crustacean issues. Vol. 3. Factors in adult growth. *Edited by* A.M. Wenner. A.A. Balkema, Rotterdam. pp. 101–128.

Haynes, E., Karinen, J.R., Watson, J., and Hopson, D.J. 1976. Relation of number of eggs and egg length to carapace width in the brachyuran crabs *Chionoecetes bairdi* and *C. opilio* from the southeastern Bering Sea and *C. opilio* from the Gulf of St. Lawrence. J. Fish. Res. Board Can. **33**: 2592–2595.

Hiatt, R.W. 1948. The biology of the lined shore crab, *Pachygrapsus crassipes* Randall. Pac. Sci. **2**: 135–213.

Hoenig, J.M., Dawe, E.G., and O Keefe, P.G. 1994. Molt indicators and growth per molt for male snow crabs (*Chionoecetes opilio*). J. Crustacean Biol. **14**: 273–279.

Hooper, R.G. 1986. A spring breeding migration of the snow crab, *Chionoecetes opilio* (O. Fabr.), into shallow water in Newfoundland. Crustaceana (Leiden), **50**: 257–264.

Ito, K. 1957. On the relative growth of abdomen and five paired thoracic appendages of Zuwaigani, *Chionoecetes opilio*. Annu. Rep. Jpn. Sea Reg. Fish. Res. Lab. **3**: 117–129.

Ito, K. 1970. Ecological studies on the edible crab, *Chionoecetes opilio* O. Fabricius in the Japan Sea. III. Age and growth as estimated on the basis of the seasonal changes in the carapace width frequencies and carapace hardness. Bull. Jpn. Sea Reg. Fish. Res. Lab. No. 22. pp. 81–116. (Translated from Japanese by Fish. Res. Board Can. Transl. Ser. No. 1512.)

Ito, K., and Kobayashi, T. 1967. A female specimen of edible crab, *Chionoecetes opilio* O. Fabricius, with the unusual symptom of additional ecdysis. Bull. Jpn. Sea Reg. Fish. Res. Lab. No. 18. pp. 127–128.

Jewett, S.C. 1981. Variations in some reproductive aspects of female snow crabs, *Chionoecetes opilio*. J. Shellfish Res. **1**: 95–99.

Kon, T. 1980. Studies on the life history of the Zuwai crab, *Chionoecetes opilio* (O. Fabricius). Spec. Publ. Sado. Mar. Biol. Stn. Niigata Univ. Ser. No. 2. (Translated from Japanese by Can. Transl. Fish. Aquat. Sci. No. 5634.)

Kurata, H. 1962. Studies on the age and growth of Crustacea. Bull. Hokkaido Reg. Fish. Res. Lab. **24**: 1–115.

Lanteigne, C., Beninger, P.G., and Gionet, C. 1996. Ontogeny of female primary sexual characters in the majid crabs *Chionoecetes opilio* and *Hyas coarctatus.* J. Crustacean Biol. **16**: 501–514.

MacDonald, P.D.M., and Pitcher, T.J. 1979. Age-groups from size-frequency data: a versatile and efficient method of analysing distribution mixtures. J. Fish. Res. Board Can. **36**: 987–1001.

Mallet, P., Conan, G.Y., and Moriyasu, M. 1993. Periodicity of spawning and duration of incubation time for *Chionoecetes opilio* in the Gulf of St. Lawrence. Int. Counc. Explor. Sea C.M. 1993/K:26.

Mauchline, J. 1976. The Hiatt growth diagram for Crustacea. Mar. Biol. **35**: 79–84.

Meusy, J.-J., and Payen, G.G. 1988. Female reproduction in malacostracan Crustacea. Zool. Sci. **5**: 217–265.

Miller, R.J. 1976. North American crab fisheries: regulations and their rationales. Fish. Bull. U.S. **74**: 623–633.

Miller, R.J., and Watson, J. 1976. Growth per molt and limb regeneration in the spider crab, *Chionoecetes opilio*. J. Fish. Res. Board Can. **33**: 1644–1649.

Moriyasu, M., Conan, G.Y., Mallet, P., Chiasson, Y., and Lacroix, H. 1987. Growth at molt, molting season and mating of snow crab (*Chionoecetes opilio*) in relation to functional maturity. Int. Counc. Explor. Sea C.M. 1987/K:21.

Paul, J.M., Paul, A.J., and Barber, W.E. 1997. Reproductive biology and distribution of the snow crab from the northeastern Chukchi Sea. *In* Fish ecology in Arctic North America. *Edited by* J. Reynolds. Am. Fish. Soc. Symp. **19**: 287–294.

Pohle, G., and Telford, M. 1982. Post-larval growth of *Dissodactylus primitivus* Bouvier, 1917 (Brachyura: Pinnotheridae) under laboratory conditions. Biol. Bull. (Woods Hole), **163**: 211–224.

Powles, H.W. 1968. Distribution and biology of the spider crab *Chionoecetes opilio* in the Magdalen Shallows, Gulf of St. Lawrence. MS Rep. Fish. Res. Board Can. No. 997.

Rice, A.L., and Hartnoll, R.G. 1983. Aspects of the biology of the deep-sea spider crab, *Dorhynchus thomsoni* (Crustacea: Brachyura). J. Zool. (London), **201**: 417–431.

Robichaud, D.A., Bailey, R.F.J., and Elner, R.W. 1989. Growth and distribution of snow crab, *Chionoecetes opilio*, in the southeastern Gulf of St. Lawrence. J. Shellfish Res. **8**: 13–23.

Sainte-Marie, B. 1993. Reproductive cycle and fecundity of primiparous and multiparous female snow crabs, *Chionoecetes opilio* (O. Fabricius), in the northwest Gulf of Saint Lawrence. Can. J. Fish. Aquat. Sci. **50**: 2147–2156.

Sainte-Marie, B., and Carrière, C. 1995. Fertilization of the second clutch of snow crab, *Chionoecetes opilio*, mated once or twice after their molt to maturity. Fish. Bull. U.S. **93**: 758–763.

Sainte-Marie, B., and Dufour, R. 1995. Crabe des neiges de l'estuaire du nord du golfe du Saint-Laurent (zones de pêche 13, 14, 15, 16 et 17). *In* Rapport sur l'état des invertébrés en 1994: crustacés et mollusques des côtes du Québec et crevette nordique de l'estuaire et du golfe du Saint-Laurent. *Edited by* L. Savard. Can. MS Rep. Fish. Aquat. Sci. No. 2323. pp. 3–27.

Sainte-Marie, B., and Hazel, F. 1992. Moulting and mating of snow crabs, *Chionoecetes opilio* (O. Fabricius), in shallow waters of the northwestern Gulf of Saint Lawrence. Can. J. Fish. Aquat. Sci. **49**: 1282–1293.

Sainte-Marie, B. and Lovrich, G.A. 1994. Delivery and storage of sperm at first mating of female *Chionoecetes opilio* (Brachyura, Majidae) in relation to size and morphometric maturity of male parent. J. Crustacean Biol. **14**: 508–521.

Sainte-Marie, B., Lapointe, V., Archambault, D., and Dufour, R. 1992. Distribution spatiale et structure de taille du crabe des neiges, *Chionoecetes opilio* (O. Fabricius), dans le fjord du Saguenay. Can. MS Rep. Fish. Aquat. Sci. No. 2156.

Sainte-Marie, B., Raymond, S., and Brêthes, J.-C. 1995. Growth and maturation of the benthic stages of male snow crab, *Chionoecetes opilio* (Brachyura: Majidae). Can. J. Fish. Aquat. Sci. **52**: 903–924.

Sainte-Marie, B., Sévigny, J.-M., Smith, B.D., and Lovrich, G.A. 1996. Recruitment variability in snow crab (*Chionoecetes opilio*): pattern, possible causes, and implications for fishery management. *In* International Symposium on Biology, Management and Economics of Crabs from High Latitude Habitats, Anchorage, Alaska, Oct. 1995. Lowell Wakefield Fish. Symp. Ser. Alaska Sea Grant Coll. Program Rep. No. 96-02. pp. 451–478.

Schafer, C.T., Smith, J.N., and Côté, R. 1990. The Saguenay Fiord: a major tributary to the St. Lawrence Estuary. *In* Oceanography of a large-scale estuarine system. The St. Lawrence. *Edited by* M.I. El-Sabh and N. Silverberg. Coastal Estuarine Stud. No. 39. pp. 378–420.

Smith, L.D. 1990. Patterns of limb loss in the blue crab *Callinectes sapidus* Rathbun, and the effect of autotomy on growth. Bull. Mar. Sci. **46:** 23–36.

Sokal, R.R., and Rohlf, F.J. 1981. Biometry. 3rd ed. W.H. Freeman & Co., San Francisco.

Somerton, D.A. 1981*a*. Life history and population dynamics of two species of Tanner crab, *Chionoecetes bairdi* and *C. opilio*, in the eastern Bering Sea with implications for the management of the commercial harvest. Ph.D. thesis, University of Washington, Seattle.

Somerton, D.A. 1981*b*. Regional variation in the size at maturity of

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two species of Tanner crab (*Chionoecetes bairdi* and *C. opilio*) in the eastern Bering Sea, and its use in defining management subareas. Can. J. Fish. Aquat. Sci. **38**: 163–174.

Somerton, D.A., and MacIntosh, R.A. 1983. The size at sexual maturity of blue king crab, *Paralithodes platypus*, in Alaska. Fish. Bull. U.S. **81**: 621–628.

Teissier, G. 1933. Étude de la croissance de quelques variants sexuels chez *Macropodia rostrata* L. Bull. Biol. France Belg. **67**: 401–444.

Teissier, G. 1935. Croissance des variants sexuels chez *Maia squinado* L. Trav. Stn. Biol. Roscoff, **13**: 93–130.

Vernet-Cornubert, G. 1958*a*. Biologie générale de *Pisa tetraodon* (Pennant). Bull. Inst. Océanogr. (Monaco) No. 1113. pp. 1–53.

Vernet-Cornubert, G. 1958*b*. Recherches sur la sexualité du crabe *Pachygrapsus marmoratus* (Fabricius). Arch. Zool. Exp. Gen. **96**: 104–274.

Watson, J. 1970. Maturity, mating and egg-laying in the spider crab, *Chionoecetes opilio*. J. Fish. Res. Board Can. **27**: 1607–1616.

Watson, J. 1972. Mating behavior in the spider crab, *Chionoecetes opilio*. J. Fish. Res. Board Can. **29**: 447–449.

Wolff, M., and Soto, M. 1992. Population dynamics of *Cancer polyodon* in La Herradura Bay, northern Chile. Mar. Ecol. Prog. Ser. **85**: 69–81.

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